

Copula Entropy

Theory and Applications

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- Lecture 7: Generalizations
- **Lecture 8: Real applications and softwares**

Lecture 8: Real applications and softwares

1 Real applications

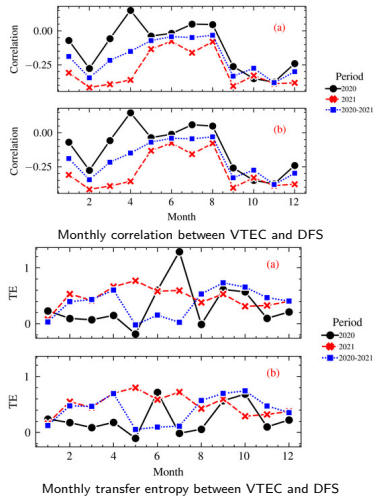
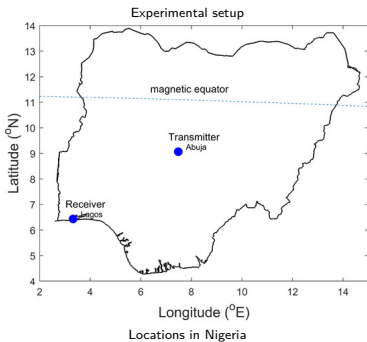
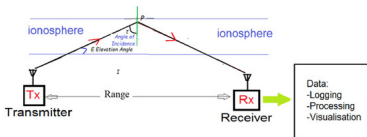
- Astronomy
- Physics
- Chemistry
- Geology
- Biology
- Medicine
- Social sciences
- Engineering

2 Softwares

- Official
- Third-party

- 赤道电离层特性随季节变化分析
 - 基于 TE 分析赤道地区高频微波信号传输中 TEC 和 DFS 之间的非线性关系³

³Aderonke Akerele et al. "Complexity and Nonlinear Dependence of Ionospheric Electron Content and Doppler Frequency Shifts in Propagating HF Radio Signals within Equatorial Regions". In: *Atmosphere* 15.6 (2024), p. 654. DOI: 10.3390/atmos15060654. URL: <https://www.mdpi.com/2073-4433/15/6/654>.

[illegible]

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流体力学

- 离心泵性能预测
 - 利用 SMOGN 过采样技术和 GAKHO 超参数调节技术相结合的方法来进行离心泵性能预测⁵

⁵Yuhao Chen et al. "Research on performance prediction of a small sample centrifugal pump based on a pre-processing approach for imbalanced regression and key hyperparameters optimization". In: *Physics of Fluids* 37.6 (2025), p. 067148. DOI: 10.1063/5.0274888. eprint: https://pubs.aip.org/aip/pof/article-pdf/doi/10.1063/5.0274888/20565541/067148_1_5.0274888.pdf. URL: <https://doi.org/10.1063/5.0274888>.

热学

- 微热管设计和制造
 - 提出了一种基于 CE 和神经网络模型相结合的微热管设计和制造参数预测方法⁶
- 循环流化床锅炉工艺参数优化
 - 结合 CE、加权融合树模型和多目标树结构估计器的循环流化床锅炉工艺参数优化方法⁷

⁶李勇 et al. “基于一维卷积残差网络的微热管几何结构及制造工艺参数预测”. In: 中国科学: 技术科学 55.2 (2025), pp. 281–294, 张靖昊. “基于机器学习的微热管结构及工艺参数设计系统开发”. 硕士学位论文. 华南理工大学, 2024.

⁷Yunpeng Ma et al. “Improve thermal efficiency and reduce NOx emission of circulating fluidized bed boiler based on multi-objective optimization framework”. In: *Thermal Science and Engineering Progress* (2025), p. 103753. ISSN: 2451-9049. DOI: <https://doi.org/10.1016/j.tsep.2025.103753>. URL: <https://www.sciencedirect.com/science/article/pii/S2451904925005438>.

- 变构效应研究
 - 变构效应配位点和激活点热力学耦合模型⁸

⁸Michel A. Cuendet, Harel Weinstein, and Michael V. LeVine. "The Allosteric Landscape: Quantifying Thermodynamic Couplings in Biomolecular Systems". In: *Journal of Chemical Theory and Computation* 12.12 (2016), pp. 5758–5767. ISSN: 1549-9618. DOI: 10.1021/acs.jctc.6b00841. URL: <https://doi.org/10.1021/acs.jctc.6b00841>.

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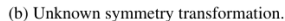
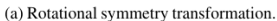


Figure 1: **Left:** a molecule is rotated by g admitting an analytical form. The distance matrix D between atoms is calculated by a known function f and remains unchanged for all rotations. **Right:** n samples $\{m, e\}^n$ where m is the molecule and e the bandgap energy. These samples approximate the function f whereas the class of functions g leading to the same bandgap energy is unknown.

¹¹ Song Zhang et al. "Correlation between plastic deformation mechanism and texture evolution of a near β -Ti alloy deformed in β region". In: *Intermetallics* 170 (2024), p. 108333. ISSN: 0966-9795. DOI: <https://doi.org/10.1016/j.intermet.2024.108333>. URL: <https://www.sciencedirect.com/science/article/pii/S0966979524001523>.

- 大气污染传播路径分析
 - 兰州市大气污染传播路径预测¹³
- 火电厂排放污染物管控
 - 火电厂氮氧化物排放浓度预测¹⁴
- 氨气污染
 - 长三角地区氨气浓度长期变化及其驱动因素¹⁵
- SO_2 排放浓度预测
 - 提出了一种 SO_2 提前预测的软测量技术¹⁶

¹³ 吴京鹏. “基于图嵌入表示的节点无特征网络链路预测研究”. 硕士学位论文. 西北师范大学, 2022.

¹⁴ 金秀章, 乔鹏, and 史德金. “基于 VMD-Bayes-Lasso 算法带误差补偿的火电厂 NO_x 浓度软测量”. In: 华北电力大学学报 (自然科学版) 52 235 (2023), pp. 117–124+142.

¹⁵ Jingkai Xue et al. "Long-term spatiotemporal variations of ammonia in the Yangtze River Delta region of China and its driving factors". In: *Journal of Environmental Sciences* 150 (2025), pp. 202–217. ISSN: 1001-0742. DOI: <https://doi.org/10.1016/j.jes.2024.02.021>. URL: <https://www.sciencedirect.com/science/article/pii/S1001074224001153>.

¹⁶ 乔鹏. “基于 CatBoost-Bayes 带误差补偿的脱硫出口 SO_2 浓度软测量研究”. 硕士学位论文. 华北电力大学, 2024.

CASE 4: Atmosphere surface coupling by Oslo University

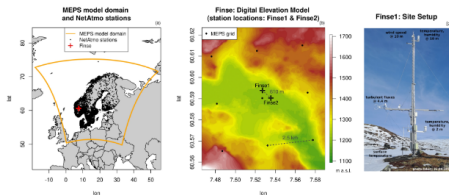


Figure 1: Used data sources: (a) Model domain of the NWP model MEPS and locations of the NetAtmo stations. (b) Surrounding of Finse and station locations. (c) Setup of the main site Finse1.

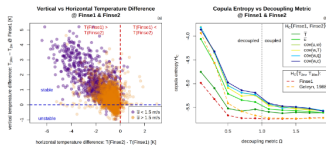
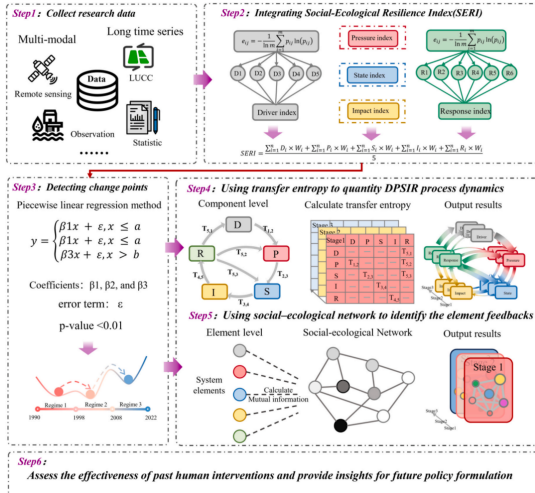


Figure 2: Relation of vertical and horizontal coupling: (a) Horizontal temperature difference between Finse1 and Finse2 versus vertical temperature difference at Finse1 colored based on a wind speed threshold (30 minutes temporal average), (b) decoupling metric at Finse1 versus copula entropy between Finse1 and Finse2 of temperature, wind speed, and the covariances corresponding to the fluxes of momentum $\text{cov}(u,w)$, sensible heat $\text{cov}(w,T)$, latent heat $\text{cov}(q,w)$ and CO_2 $\text{cov}(c,w)$ (30 minutes temporal average). Entropy calculation is based on time series of the same length to avoid a sampling size bias.

CASE 7: Social-Eco-Resilience in Karst region by CAS



Research methodology with copula entropy based transfer entropy

神经科学 1

● 神经病学

- 注意缺陷多动障碍⁶⁰
- 癫痫症⁶¹
- 脑卒中⁶²

⁶⁰ Paolo Victor Redondo, Raphaël Huser, and Hernando Ombao. "Measuring information transfer between nodes in a brain network through spectral transfer entropy". In: *The Annals of Applied Statistics* 19.3 (2025), pp. 2386–2411. DOI: [10.1214/25-AOAS2050](https://doi.org/10.1214/25-AOAS2050). URL: <https://doi.org/10.1214/25-AOAS2050>, 汪方毅 et al. "基于静息态 fMRI 区分健康老年人认知水平的 MVPA 方法研究". In: *磁共振成像* 14.6 (2023), pp. 18–25.

⁶¹ Zhaohui Li et al. "Estimating global phase synchronization by quantifying multivariate mutual information and detecting network structure". In: *Neural Networks* 183 (2025), p. 106984. ISSN: 0893-6080. DOI: <https://doi.org/10.1016/j.neunet.2024.106984>. URL: <https://www.sciencedirect.com/science/article/pii/S0893608024009134>.

⁶² Wojciech Ciezobka et al. "End-to-End Stroke Imaging Analysis Using Effective Connectivity and Interpretable Artificial Intelligence". In: *IEEE Access* 13 (2025), pp. 10227–10239. DOI: [10.1109/ACCESS.2025.3529179](https://doi.org/10.1109/ACCESS.2025.3529179).

神经科学 2

● 认知神经学

- 分析大脑认知活动的多模态数据⁶³
- 语音信息的编码和解析⁶⁴
- 神经元特异化识别⁶⁵

⁶³Stephanie J. Kayser et al. "Irregular Speech Rate Dissociates Auditory Cortical Entrainment, Evoked Responses, and Frontal Alpha". In: *The Journal of Neuroscience* 35.44 (2015), pp. 14691–14701, Robin A. A. Ince et al. "The Deceptively Simple N170 Reflects Network Information Processing Mechanisms Involving Visual Feature Coding and Transfer Across Hemispheres". In: *Cerebral Cortex* 26.11 (2016), pp. 4123–4135, Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Etienne Combrisson et al. "Group-level inference of information-based measures for the analyses of cognitive brain networks from neurophysiological data". In: *NeuroImage* 258 (2022), p. 119347. ISSN: 1053-8119. DOI: <https://doi.org/10.1016/j.neuroimage.2022.119347>. URL: <https://www.sciencedirect.com/science/article/pii/S1053811922004669>.

⁶⁴Pieter De Clercq et al. "Beyond linear neural envelope tracking: a mutual information approach". In: *Journal of Neural Engineering* 20.2 (2023), p. 026007. DOI: 10.1088/1741-2552/acbe1d. URL: <https://dx.doi.org/10.1088/1741-2552/acbe1d>.

⁶⁵NA Pospelov et al. "Searching for cognitive specializations of neurons using mutual information framework". In: *Genes & Cells* 18.4 (2023), pp. 878–881.

心理学

- 生物心理学
 - 情绪刺激下心跳诱发脑电位的时间交互现象⁶⁹

⁶⁹ Liesa Ravijts. "Revealing temporal interactions around the heartbeat-evoked potential modulated by emotional perception". MA thesis. Ghent Univeristy, 2019.

医学 2

● 临床医学 2

- 白内障术后角膜水肿风险预测⁷⁸
- 主动脉瓣置换手术射血分数分析⁷⁹
- 脑肿瘤医学影像组学诊断模型构建⁸⁰
- 基于脉搏波的高血压和糖尿病健康状态分类⁸¹

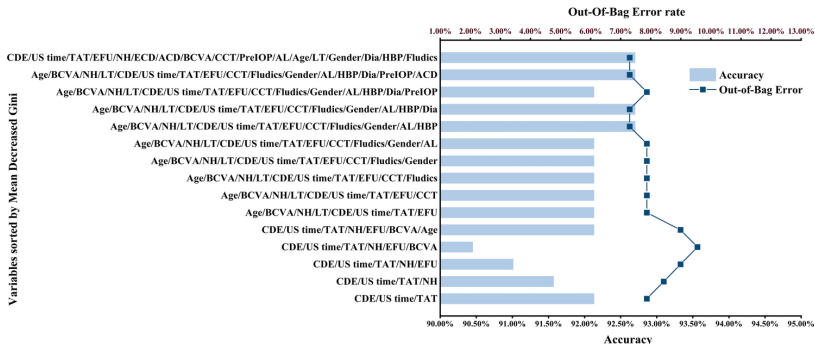
⁷⁸Yu Luo et al. "Research on Establishing Corneal Edema after Phacoemulsification Prediction Model Based on Variable Selection with Copula Entropy". In: *Journal of Clinical Medicine* 12.4 (2023), p. 1290. ISSN: 2077-0383. DOI: 10.3390/jcm12041290. URL: <https://www.mdpi.com/2077-0383/12/4/1290>.

⁷⁹S.M. Sunoj and N. Unnikrishnan Nair. "Survival Copula Entropy and Dependence in Bivariate Distributions". In: *REVSTAT-Statistical Journal* 23.1 (2025), pp. 101–115. URL: <https://revstat.ine.pt/index.php/REVSTAT/article/view/560>.

⁸⁰潘红宇. "基于影像组学与深度学习的脑肿瘤图像分类研究". 硕士学位论文. 西南大学, 2023.

⁸¹汤宇飞. "基于脉搏波的糖尿病和高血压诊断算法研究". 硕士学位论文. 中国矿业大学, 2023.

CASE 10: Corneal edema prediction by PLA 301 Hospital



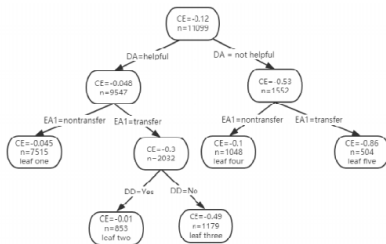
The predictive accuracy and out-of-bag error rate of each random forest model generated after selecting variables on the basis of copula entropy

药学

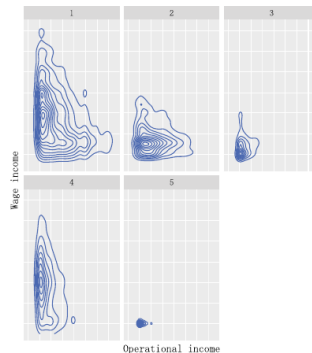
- 药物-靶点相互作用预测
 - 提出了一种交互式多特征融合的 DTIs 预测算法⁸⁷

⁸⁷ 高浩田 et al. “基于交互式多特征融合算法的药物靶标预测”. In: 太原理工大学学报 55.4 (2024), pp. 751–758.

CASE 11: Poverty policy evaluation by JXUFE



Binary decision tree by copula entropy



Joint density of two incomes of the families in each leaf

CASE 12: Commodity market analysis by UNESP



Figura 5.0.5: Transfer Entropy entre as séries da volatilidade de preços do açúcar e etanol.



Figura 5.0.6: Transfer Entropy entre as séries da volatilidade de preços do açúcar e gasolina.



Figura 5.0.7: Transfer Entropy entre as séries da volatilidade de preços do etanol e gasolina.

CASE 13: LLMs evaluation by Cambridge University

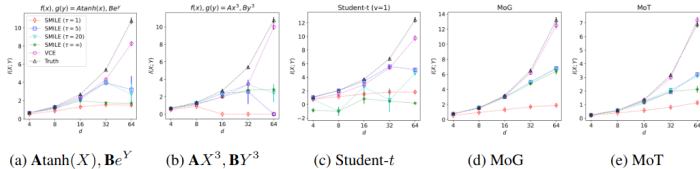


Figure 8: Comparison with the SMILE estimator under different clipping values τ .

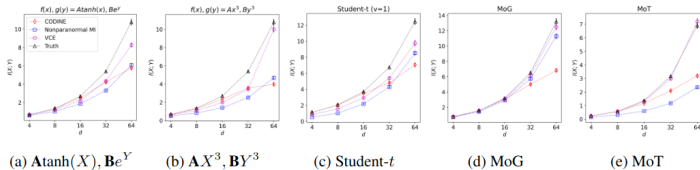


Figure 9: Comparing VCE with classic copula-based estimators e.g. nonparanormal MI (which uses a Gaussian copula to estimate MI) and CODINE (equivalent to classic copula transformation + MINE).

社会科学 4

- 法学
 - 社区属性与社区犯罪关系分析¹⁰⁵
- 政治学
 - 分析政权领导力因素和政权危机之间关系¹⁰⁶
- 军事学
 - 目标意图识别¹⁰⁷
- 情报学
 - 颠覆性技术科学-技术-产业互动模式分析¹⁰⁸
 - 高价值专利评估和发现¹⁰⁹

¹⁰⁵ Mario Wieser. "Learning Invariant Representations for Deep Latent Variable Models". PhD thesis. University of Basel, 2020.

¹⁰⁶ Stuart William Card. "Towards an Information Theoretic Framework for Evolutionary Learning". PhD thesis. Syracuse University, 2011.

¹⁰⁷ 张可 et al. "一种基于动态贝叶斯网络的目标意图识别方法". Pat. CN114997306A. CN114997306A. 2022.

¹⁰⁸ 许海云 et al. "颠覆性技术的科学-技术-产业互动模式识别与分析". In: 情报学报 42.7 (2023), pp. 816–831.

¹⁰⁹ 徐蕊婷, 陈哲宇, and 陈加鹏. "基于两阶段混合式特征选择的潜在高价值专利多情景判别方法研究". In: 数据分析与知识发现 (2026).

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CASE 14: Cyber-Physical integration by National Grid

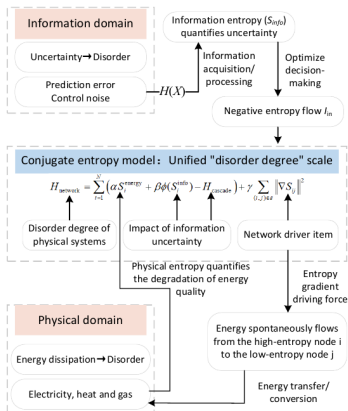


Fig. 1. The core concept diagram of the conjugate entropy theory.

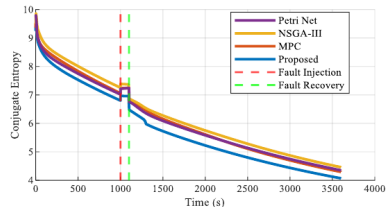


Fig. 7. Comparison of conjugate entropy.

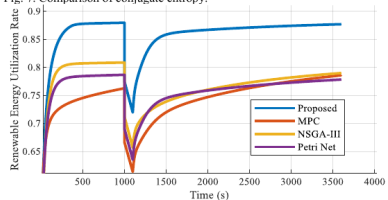


Fig. 8. Comparison of renewable energy utilization rate.

工学 4

- 纺织工程
 - 纱布生产过程质量控制¹²³
 - 可穿戴织物传感器方案设计¹²⁴
- 食品工程
 - 葡萄酒质量与理化成分关系分析¹²⁵

¹²³Seyedeh Azadeh Fallah Mortezaejad et al. "Non-Parametric Multivariate Control Chart Using Copula Entropy". In: *Sankhya B* 87 (2025), pp. 468–518. URL: <https://doi.org/10.1007/s13571-025-00374-y>.

¹²⁴Jiaqi Mo et al. "An Evaluation Framework for Textile Stretch Sensor Layouts Based on Mutual Information". In: *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. UbiComp Companion '25. Finland: Association for Computing Machinery, 2026, pp. 35–39. ISBN: 9798400714771. DOI: 10.1145/3714394.3754434. URL: <https://doi.org/10.1145/3714394.3754434>.

¹²⁵Marvin Lasserre, Régis Lebrun, and Pierre-Henri Willemin. "Learning Continuous High-Dimensional Models using Mutual Information and Copula Bayesian Networks". In: *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021*. AAAI Press, 2021, pp. 12139–12146. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/17441>, Marvin Lasserre. "Apprentissages dans les réseaux bayésiens à base de copules non-paramétriques". PhD thesis. Sorbonne Université, 2022. URL: <https://tel.archives-ouvertes.fr/tel-03647090>.

CASE 15: Wine quality by Sorbonne Université

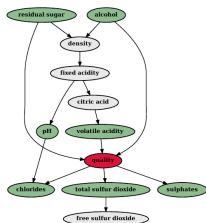


FIGURE 8.10 – Structure reconstruite avec l'algorithme B-CMIIC à partir de l'échantillon des vins rouges. Le paramètre α a été fixé à 0.04 en utilisant la méthode de *cross-validation*. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.



FIGURE 8.11 – Structure reconstruite avec l'algorithme B-CMIIC pour l'échantillon des vins blancs. Le paramètre α est fixé à 0.02. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.



FIGURE 8.12 – Structure reconstruite avec l'algorithme B-CMIIC pour l'échantillon des vins blancs. Le paramètre α est fixé à 0.06. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.

工学 10

- 医学工程
 - 脑机接口¹⁴⁹
- 冶金工程
 - 真空蒸馏法制备高纯金属工艺参数优化¹⁵⁰

¹⁴⁹Chao Tang, Dongyao Jiang, and Badong Chen. "MEG Channel Selection Using Copula Entropy-Based Transfer Entropy for Motor Imagery BCI". In: *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 2024, pp. 1–4. DOI: 10.1109/EMBC53108.2024.10782066, Chao Tang et al. "Copula Transfer Entropy-Based Channel Selection for MEG Motor Imagery Brain Computer Interfaces". In: *Tsinghua Science and Technology* 31.3 (2026), pp. 1474–1486. DOI: 10.26599/TST.2025.9010006. URL: <https://www.sciopen.com/article/10.26599/TST.2025.9010006>.

¹⁵⁰田庆华 et al. "一种真空蒸馏制备高纯金属的优化方法及优化系统". Pat. CN117577229A. CN117577229A. 2024.

工学 11

● 化学工程

- 化学过程故障监测和诊断¹⁵¹
- 化工过程因果网络构建¹⁵²
- 化工过程缺失数据补全¹⁵³
- 生物发酵过程软测量¹⁵⁴

¹⁵¹Min Yin, Jince Li, and Hongguang Li. "A CNN approach based on correlation metrics to chemical process fault classifications with limited labeled data". In: *The Canadian Journal of Chemical Engineering* 101.7 (2022), pp. 3982–3997. doi: 10.1002/cjce.24749. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/cjce.24749>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/cjce.24749>, Yingpeng Wei and Li Wang. "Copula entropy-based PCA method and application in process monitoring". In: *2022 4th International Conference on Intelligent Information Processing (IIP)*. 2022, pp. 61–64. DOI: 10.1109/IIP57348.2022.00019, Shuangshuang Pan, Li Zhu, and Xirong Xu. "Root cause and fault propagation analysis based on causal graph in chemical processes". In: *2023 CAA Symposium on Fault Detection, Supervision and Safety for Technical Processes (SAFEPROCESS)*. 2023, pp. 1–6. DOI: 10.1109/SAFEPROCESS58597.2023.10295717.

¹⁵²Xiaotian Bi et al. "Large-scale chemical process causal discovery from big data with Transformer-based deep learning". In: *Process Safety and Environmental Protection* 173 (2023), pp. 163–177.

¹⁵³武昊. "基于深度学习的化工过程软测量建模方法研究". 博士学位论文. 北京化工大学, 2023.

¹⁵⁴时和畅. "基于时序差分神经网络的发酵过程批次终点预测及优化研究". 硕士学位论文. 北京化工大学, 2025.

工学 12

- 核工程
 - 核反应堆传感器故障与事故预防¹⁵⁵

155 邓志光 et al. “铅铋快堆多工况多传感器状态监测方法、装置和验证系统”. Pat. CN121350688A. CN121350688A. 2026, Jiarong Gao et al. “Research on nuclear power plant sensor data reconstruction based on Seq2Seq-Mamba”. In: *Annals of Nuclear Energy* 231 (2026), p. 112219. ISSN: 0306-4549. DOI: <https://doi.org/10.1016/j.anucene.2026.112219>. URL: <https://www.sciencedirect.com/science/article/pii/S0306454926001076>.

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金融工程 1

- 投资组合优化
 - 股票资产相关性网络分析¹⁷⁶
 - ST 股票分类¹⁷⁷
- 金融问题建模
 - Copula 函数模型选择¹⁷⁸
- 股票相关性建模
 - R-vine copula 结构建模¹⁷⁹

¹⁷⁶ Qitong Wang. "Social Networks, Asset Allocation and Portfolio Diversification". MA thesis. University of Waterloo, 2015, 廖轶楠. "基于 Copula 熵选股及集成神经网络预测的投资组合管理研究". 硕士学位论文. 南京信息工程大学, 2023.

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¹⁷⁹ Fadhah Amer Alanazi. "Truncating Regular Vine Copula Based on Mutual Information: An Efficient Parsimonious Model for High-Dimensional Data". In: *Mathematical Problems in Engineering* 2021 (2021), p. 4347957. URL: <https://doi.org/10.1155/2021/4347957>, 王念鸽. "基于互信息的 Vine Copula 模型的高频数据投资组合风险测度研究". 硕士学位论文. 浙江财经大学, 2023.

金融工程 2

- 量化金融工具箱 MLFinLab¹⁸⁰
- 金融系统性风险
 - 行业风险溢出效应分析¹⁸¹
 - 金融脆弱性度量¹⁸²
- 信用风险评价
 - 信用风险卡模型建立¹⁸³
 - P2P 借贷个人信用风险模型¹⁸⁴

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¹⁸¹熊靖宇. “基于 Copula 熵的行业风险溢出效应分析”. 硕士学位论文. 东北财经大学, 2020.

¹⁸²Wanfang Chen and Marc G Genton. “Are you all normal? It depends!” In: *International Statistical Review* 91.1 (2023), pp. 114–139.

¹⁸³孔祥永 et al. “一种自动化特征工程信用风险评价系统及方法”. Pat. CN114049198A. CN114049198A. 2021.

¹⁸⁴彭翊庭. “个人信用风险评估模型比较——基于 Copula 熵的特征选择”. 硕士学位论文. 清华大学, 2022.

Softwares

Third-party

- R: `cylcop`¹⁸⁹ (by University of Zürich and University of Minnesota);
- Python: `MLFinLab`¹⁹⁰ and `ArbitrageLab`¹⁹¹ (authored by Prof. Marcos López de Prado from Cornell University, Github 4500+ and 670+ stars);

¹⁸⁹Florian H. Hodel and John R. Fieberg. "cylcop: An R Package for Circular-Linear Copulae with Angular Symmetry". In: *bioRxiv* (2021), p. 2021.07.14.452253, Florian Hodel. *cylcop: Circular-Linear Copulas with Angular Symmetry for Movement Data*. CRAN. R package version 0.2.0, URL: <https://cran.r-project.org/package=cylcop>. 2022.

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Softwares

Third-party

- Python: **gcmi**¹⁹² (by Glasgow University, Radboud University Nijmegen, University of Amsterdam, Kyiv AI group, Github 100+ stars), **pytorch-mighty**¹⁹³ (by Kyiv AI group), **HOI**¹⁹⁴ and **Frites**¹⁹⁵ (by CNRS, INT, and Aix-Marseille Université), **Tensorpac**¹⁹⁶ (by Université de Montréal, INT, and Aix-Marseille Université), **driada**¹⁹⁷ (by Moscow State University);

¹⁹²Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Robin A.A. Ince et al. *gcmi* : *Gaussian-Copula Mutual Information*. GitHub. URL: <https://github.com/robinince/gcmi>. 2020.

¹⁹³Danylo Ulianych. *The pytorch-mighty package in Python*. GitHub. 2023. URL: <https://github.com/dizcza/pytorch-mighty>.

¹⁹⁴BraiNets. *The HOI package in Python*. Github. 2024. URL: <https://brainets.github.io/hoi/>.

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¹⁹⁷Institute for Advanced Brain Studies. *driada: Dimensionality Reduction for Integrated Activity Data*. Github. URL: <https://github.com/iabs-neuro/driada>. 2024.

Softwares

Third-party

- Python: CopulaGP¹⁹⁸ (by University of Edinburgh), **Polars-ds**¹⁹⁹ (by New York University, 1.45+ million downloads, Github 650+ stars), **copul**²⁰⁰ (by Albert-Ludwigs-Universität Freiburg), **rcopula**²⁰¹, and **pycopcor**²⁰² (by Technische Universität Dresden);

¹⁹⁸Nina Kudryashova et al. "Parametric Copula-GP model for analyzing multidimensional neuronal and behavioral relationships". In: *PLoS computational biology* 18.1 (2022), e1009799, Nina Kudryashova. *Parametric Copula-GP framework*. Github. URL: <https://github.com/NinelK/CopulaGP>. 2022.

¹⁹⁹Tianren Qin. *Polars-ds: Polars Extension for General Data Science Use*. Github. URL: https://github.com/abstractqqq/polars_ds_extension. 2024.

²⁰⁰Marcus Rockel. *The copul package in Python*. Github. URL: <https://github.com/Corrram/copul>. 2026.

²⁰¹Christopher Bell. *The rcopula package in Python*. Github. URL: <https://github.com/Chrisebell124/python-copula>. 2026.

²⁰²Andreas Gocht-Zech. *The pycopcor package in Python*. Github. URL: <https://github.com/AndreasGocht/pycopcor>. 2023.

Third-party

- **Julia:** `CopEnt.jl`²⁰³ (by University of Edinburgh), `CausalityTools.jl`²⁰⁴ (by JuliaDynamics), and `Copulas.jl`²⁰⁵ (by Aix-Marseille Université, Github 110+ stars);
- **Matlab:** `gcmi`²⁰⁶ (by Glasgow University, Radboud University Nijmegen, University of Amsterdam, Kyiv AI group, Github 100+ stars), and `FieldTrip`²⁰⁷ (by Radboud University Nijmegen);
- **C++:** `NPStat`²⁰⁸ ²⁰⁹ (by Texas Tech University).

²⁰³Kai Xu. *The CopEnt.jl package in Julia*. Github. 2021. URL: <https://github.com/xukai92/CopEnt.jl>.

²⁰⁴Kristian Agasøster Haaga. *The CausalityTools.jl package v2.10.1 in Julia*. Github. 2023. URL: <https://github.com/JuliaDynamics/CausalityTools.jl>.

²⁰⁵Oskar Lavrny and Santiago Jimenez. "Copulas.jl: A fully Distributions.jl-compliant copula package". In: *Journal of Open Source Software* 9.94 (2024), p. 6189. DOI: 10.21105/joss.06189. URL: <https://doi.org/10.21105/joss.06189>.

²⁰⁶Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Robin A.A. Ince et al. *gcmi* : Gaussian-Copula Mutual Information. GitHub. URL: <https://github.com/robinince/gcml>. 2020.

²⁰⁷ Robert Oostenveld et al. "FieldTrip: Open Source Software for Advanced Analysis of MEG, EEG, and Invasive Electrophysiological Data". In: *Computational Intelligence and Neuroscience* 2011.1 (2011), p. 156869. DOI: <https://doi.org/10.1155/2011/156869>.

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²⁰⁹The CE estimation method is decomposed into two functions for empirical copula estimation and entropy estimation.

Thanks!

Q&A